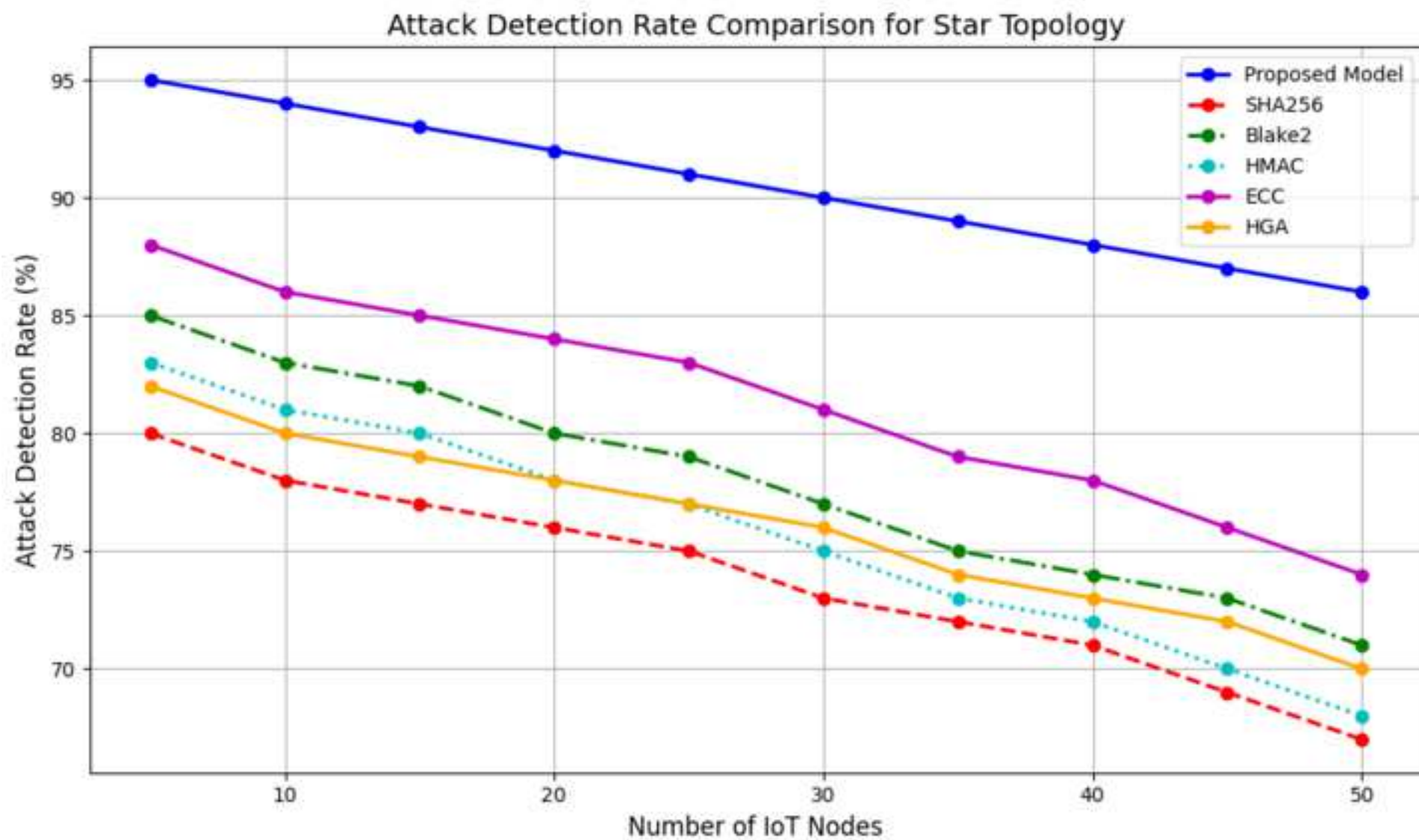


# Arabian Journal for Science and Engineering

## ECP-SCA: Enhanced Curve448 Pre-Computation for Side-Channel Attack Mitigation in UAV Network --Manuscript Draft--

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Manuscript Number:</b>                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Full Title:</b>                                   | ECP-SCA: Enhanced Curve448 Pre-Computation for Side-Channel Attack Mitigation in UAV Network                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Article Type:</b>                                 | SCI / ENG - Research Article                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Section/Category:</b>                             | ENG-Computer Engineering and Computer Science                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Keywords:</b>                                     | Cybersecurity, Digital infrastructure; Unmanned Aerial Vehicles (UAVs); Elliptic Curve Cryptography (ECC); Side-Channel Attacks, Secure UAV Communication; Key Exchange Protocol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Corresponding Author:</b>                         | Lokesh Sharma, Ph.D., M.Tech., B.E.<br>Manipal University Jaipur<br>Jaipur, Rajasthan INDIA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Corresponding Author Secondary Information:</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Corresponding Author's Institution:</b>           | Manipal University Jaipur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Corresponding Author's Secondary Institution:</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>First Author:</b>                                 | Mahesh Jangid, Ph.D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>First Author Secondary Information:</b>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Order of Authors:</b>                             | Mahesh Jangid, Ph.D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                      | Lokesh Sharma, Ph.D., M.Tech., B.E.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                      | Inderjeet Kaur, Ph.D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                      | Prashant Hemrajani, Ph.D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Order of Authors Secondary Information:</b>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Funding Information:</b>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Abstract:</b>                                     | <p>The increasing use of Unmanned Aerial Vehicles (UAVs) in critical applications has created a strong need for secure communication systems. UAV networks are vulnerable to cyber threats such as unauthorized access, data tampering, information leakage, and interception of sensitive communication, which can affect both data integrity and network reliability. In addition, side-channel attacks further increase the security challenges in UAV communication systems.</p> <p>This paper proposes an efficient authentication framework based on Fixed-Base Window Precomputation (FBWP) using the Curve448 elliptic curve. The proposed method improves the efficiency of cryptographic operations in Elliptic Curve Cryptography (ECC) by precomputing and storing multiples of a base point, thereby reducing computation time and energy consumption during scalar multiplication. Curve448 provides a high security level of 224 bits, offering strong resistance against modern cryptographic and side-channel attacks while maintaining computational efficiency.</p> <p>The proposed FBWP approach is integrated with the Elliptic Curve Diffie-Hellman (ECDH) protocol to enable secure key exchange in UAV communications. Experimental results demonstrate significant improvements in resistance against timing attacks, power analysis attacks, and electromagnetic leakage attacks. The proposed model achieves approximately 90% resistance to side-channel attacks and enhances both the security and reliability of UAV communication networks.</p> |
| <b>Suggested Reviewers:</b>                          | Chhagan Lal, Ph.D<br>Postdoctoral Fellow, Norwegian University of Science and Technology: Norges teknisk-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>naturvitenskapelige universitet<br/>chhagan.lal@ntnu.no</p> <p>The reviewer has over 8 years of experience in cybersecurity, privacy, and cloud security, with expertise in security frameworks, risk assessment, threat intelligence, and machine learning applications in cybersecurity. Currently associated with the CISaR group at NTNU and contributing to the EU-funded CYBERUNITY project, he possesses strong research and technical expertise. His internationally recognized certifications, including CISSP, CCSP, and Microsoft Azure Security certifications, further demonstrate his competence and suitability as a reviewer for high-quality journal publications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | <p>Vipin Pal, Ph.D<br/>Assistant Professor, National Institute of Technology Delhi<br/>vipinpal@nitdelhi.ac.in</p> <p>Dr. Vipin Pal is currently serving as Assistant Professor (Grade-I) in the Department of Computer Science and Engineering at the National Institute of Technology (NIT) Delhi. He completed his Ph.D. from Malaviya National Institute of Technology (MNIT) Jaipur and has previously served at NSUT New Delhi and NIT Meghalaya. His research expertise includes Wireless Sensor Networks, Data Mining, Text Mining, Machine Learning, IoT, Security, and Computer Networks. With strong academic and research experience in emerging technologies and interdisciplinary domains, he possesses the required technical expertise and analytical skills for critically evaluating research manuscripts. His teaching, research, and professional contributions make him highly suitable as a reviewer for quality journal publications.</p>                                                                                                                                                                                                                                                                                                                                                 |
|  | <p>Sarada Gochhayat, Ph.D<br/>Professor, Indian Institute of Technology Jammu<br/>sarada.gochhayat@iitjammu.ac.in</p> <p>Dr. Sarada Prasad Gochhayat is currently serving as an Assistant Professor at the Indian Institute of Technology (IIT) Jammu. He holds a Ph.D. in Communication Engineering from the Indian Institute of Science (IISc) Bangalore and possesses extensive academic, teaching, and research experience in cybersecurity, blockchain, cloud computing, IoT security, network security, and privacy-aware distributed systems. He has previously served at reputed institutions including Villanova University (USA), Old Dominion University (USA), Manipal University Jaipur, and CV Raman Global University.</p> <p>His research contributions span secure and privacy-preserving protocols, blockchain-based systems, physical layer security, and cyber-physical systems. He has also contributed as a Guest Editor for reputed indexed journals and has been actively involved in international research collaborations and projects. With his strong interdisciplinary expertise, research background, and extensive teaching experience in computer networks and cybersecurity-related domains, he is highly suitable to serve as a reviewer for quality journal publications.</p> |
|  | <p>karan Verma, Ph.D<br/>Associate Professor, National Institute of Technology Delhi<br/>karanverma@nitdelhi.ac.in</p> <p>Dr. Karan Verma is currently serving as an Associate Professor in the Department of Computer Science and Engineering at the National Institute of Technology (NIT) Delhi. He possesses extensive academic and research experience in emerging areas such as Ubiquitous Computing, Federated Learning, Deep Learning, 5G/6G Communication, Cloud Computing, Next-Generation Networks, Space–Terrestrial Integrated Networks, and Internet of Things (IoT).</p> <p>With strong expertise in advanced computing and communication technologies, Dr. Verma has actively contributed to research, teaching, and the promotion of innovation in computer science and engineering. His interdisciplinary research background, technical proficiency, and analytical capabilities make him highly suitable for evaluating research manuscripts in cutting-edge domains. Considering his academic contributions and specialization in emerging technologies, he is well-qualified to serve as a reviewer for quality journal publications.</p>                                                                                                                                                  |



# ECP-SCA: Enhanced Curve448 Pre-Computation for Side-Channel Attack Mitigation in UAV Network

Mahesh Jangid<sup>1†</sup>, Lokesh Sharma<sup>2\*†</sup>, Inderjeet Kaur<sup>3†</sup>,  
Prashant Hemrajani<sup>4†</sup>

<sup>1</sup>Department of Computer Science and Engineering, Manipal University  
Jaipur, Dehmi Kalan, Jaipur, 303007, Rajasthan, India.

<sup>2\*</sup>Department of Information Technology, Manipal University Jaipur,  
Dehmi Kalan, Jaipur, 303007, Rajasthan, India.

<sup>3</sup>Department of Computer Science and Engineering, Ajay Kumar Garg  
Engineering College, , Ghaziabad, India.

<sup>4</sup>Department of Internet of Things & Intelligent Systems, Manipal  
University Jaipur, Dehmi Kalan, Jaipur, 303007, Rajasthan, India.

\*Corresponding author(s). E-mail(s):

[lokesh.sharma@jaipur.manipal.edu](mailto:lokesh.sharma@jaipur.manipal.edu);

Contributing authors: [mahesh.jangid@jaipur.manipal.edu](mailto:mahesh.jangid@jaipur.manipal.edu);  
[kaurinderjeet@akgec.ac.in](mailto:kaurinderjeet@akgec.ac.in); [prashant.hemrajani@jaipur.manipal.edu](mailto:prashant.hemrajani@jaipur.manipal.edu);

<sup>†</sup>These authors contributed equally to this work.

## Abstract

The increasing use of Unmanned Aerial Vehicles (UAVs) in critical operations has made evident the need for strong security features to protect UAV communication networks. These networks are vulnerable to several possible attacks, such as data breaches, unauthorised data tampering and interception of sensitive information during communications, which leads to compromised data integrity as well as network resilience. However, the difficulty of protecting UAV networks is complicated by the increased complexity of side-channel attacks. In this paper, we propose an optimal authentication system based on Fixed-Base Window Pre-computation (FBWP) using the Curve448 elliptic curve, which has improved the operational efficiency of cryptographic operations used in Elliptic Curve Cryptography (ECC). The FBWP method allows for precomputation and storage of multiples of a base point so that the computational latency and the energy

consumption of scalar multiplication (a vital part of ECC cryptologic operations) is ameliorated. Additionally, the use of Curve448 provides 224-bits security level, which affords a great resistance to contemporary cryptologic attacks and side-channel attacks while maintaining computational efficiency. The proposed adaptation of FBWP is combined with the Elliptic Curve Diffie-Hellmann (ECDH) protocol to ensure secure key exchange in UAV communication. Results of evaluation demonstrate a significant increase in the improvement of performance variables such as power analysis, electromagnetic leakage and timing attack resistance etc. for the effective amelioration of side-channel attacks. The proposed model achieves a resistance factor 90% to side-channel attacks in the case of UAV communication and shows a significant improvement in both the security and reliability of UAV networks.

**Keywords:** Cybersecurity, Digital infrastructure, Unmanned Aerial Vehicles (UAVs), Elliptic Curve Cryptography (ECC), Side-Channel Attacks, Secure UAV Communication, Key Exchange Protocol

## 1 Introduction

Unmanned Aerial Vehicles (UAVs) [1], or drones, are aircraft controlled by a radio remote control and/or autonomous computer programmed systems. Their small size, cheapness and versatility make them important in various fields such as geophysical surveys, traffic control, film production, homeland security, and reconnaissance on the battlefield. The UAVs have an important function of surveillance, or reconnaissance, or search and rescue, or survey, as well as monitoring and doing logistics for both military and civilian interests [2].

However, with the increasing economic importance and commonality of the UAVs, security of its wireless data communication systems is of supreme concern [3]. Because of the high-risk missions on which the UAVs are employed, e.g. sensitive data communication, these vehicles have become the target of choice for hackers and other cybercriminals and renegade elements. Hence, a strong encryption mechanism is a mandatory necessity to ensure a secure communication process, for otherwise the breach of data information systems would defeat the purpose of the missions and make them a danger to public safety. But there are threats to privacy and national security in the possible cyber-nefarious implementations of UAVs, e.g. espionage, unauthorised surveillance, etc. [4].

One of the most precarious vulnerabilities of UAV systems lies in the susceptibility to side-channel attacks. Side-channel attacks exploit the result of the accidental leakage of such physical emissions as power consumption, electromagnetic emissions, and transit times to infer sensitive information during cryptographic operations. Such attacks make it unnecessary to circumvent normal safeguards, with the result that cryptographic keys and other forms of secure data are compromised [5]. The situation is much aggravated where wireless communications are employed because the facility to be attacked by interception, tampering or denial-of-service attacks is present.

While the various symmetric and asymmetric cryptographic protocols lend themselves to the security of UAV communications as a whole, the computational complexity and energy overhead required in these protocols make them unsuitable in most instances to the resource-constrained systems of UAVs. Symmetric cryptography such as the Advanced Encryption Standard (AES) [6] is clearly strong encryption but has an overhead that prevents UAVs from being used in real-time situations, especially where high data rates are present. On the other hand, Elliptic Curve Cryptography (ECC) [7] introduced by Koblitz and Miller has provided strong security with the small key size suited to UAV systems. Its great efficiency in resource usage, when taken into account with high security level means that it is extensively used in embedded systems for the solution of key management and digital signatures [8].

Although elliptic curve cryptography (ECC) has its implementation challenges. The modular arithmetic associated with ECC operations is costly in terms of the limited computing power capability of UAV processors. Furthermore, the UAVs are subject to attack by side channels, such as Simple Power Analysis (SPA) and Differential Power Analysis (DPA). Curve25519 proposed by Bernstein, was particularly noted for its amelioration of ECC against SPA and timing attacks and maintained a security level equal to AES-128 [9]. However, although some improvements in this area have been achieved, there are still some weaknesses. For example, key recovery attacks targeting Fixed-Base Comb (FBC) scalar multiplication ECC operations have revealed mathematical weaknesses that can significantly reduce the effort necessary in order to recover cryptographic keys [10].

As a result of these challenges, there is a continuing requirement for cryptographic means which are both safer and less computationally expensive to apply to UAV applications. It will be the objective of this research paper to look into improved methods by employing an alternative to Curve-25519, viz., Curve448 and the Fixed-Base Window Precomputation technique for the efficient scalar multiplication. Improved performance and efficiency in the scalar multiplication function, plus the more efficient immunity from side-channel attacks associated with the technique used for performing it, will prove to greatly enhance the security of UAV networks without compromising operational performance. Thus, the approach makes it well-suited for real-world deployments. The results of this paper will aid in developing cutting-edge technology and will also help to advance UAV security, specifically within resource-limited and high-risk areas.

The rapid proliferation of Unmanned Aerial Vehicles (UAVs) in military, business and civilian realms has brought significant advancements, but it has also raised critical security issues. Of these security attacks, side channel attacks (SCAs) are a major attack on UAV networks. SCAs exploit physical leakages like power consumption, timing information, electromagnetic emissions, etc. to exploit the cryptographic keys, which leads to compromising sensitive operations and gives more opportunities for security breaches.

Classic cryptographic tools, despite providing good security, are inadequate for UAV environments due to their high processing power and energy demands, as UAVs have limited computing and energy resources. They require processing power, memory, and an energy limit, which totally makes classical approaches unacceptable. Therefore,



there is a need for suitable and efficient cryptographic schemes to cope with the side-channel attack problem while making the best use of resources to achieve good and secure UAV communications.

This research addresses the challenging task of securing UAV communications against side-channel attacks while simultaneously optimising resource use. The main contributions of this work are summarised in the following:

- **Efficient authentication using Curve448:** A novel Fixed-Base Window Pre-computation technique based on the Curve448 elliptic curve is introduced. This allows scalar multiplication, a fundamental operation in Elliptic Curve Cryptography (ECC), to be performed more efficiently, enhancing both the security and the efficiency of UAV communications.
- **Reduction of Side-Channel Attacks (SCAs):** The proposed system uses optimised pre-computation techniques to reduce the sensitive data leakage in scalar multiplication procedures. This decreases the susceptibility of UAV network communication to SCAs such as power analysis, timing attacks, and exploitation of electromagnetic emission. Additionally, it ensures efficient memory usage with increasing resource efficiencies.
- **Extensive security evaluation:** An in-depth security assessment demonstrated the capability of the proposed technique to resist various side-channel attack techniques such as differential power analysis (DPA) and electromagnetic emanation analysis. The assessment validates the inherent strength of the system in combating side-channel attacks while maintaining efficient computational management.

The remainder of this paper is organized as follows. Section 2 reviews the related work on UAV communication security, elliptic curve cryptography, and existing approaches for mitigating side-channel attacks. Section 3 presents the proposed methodology, describing the ECP-SCA framework based on Fixed-Base Window Precomputation (FBWP) using Curve448 for secure and efficient authentication in UAV networks. Section 4 details the Montgomery Ladder algorithm for efficient and constant-time scalar multiplication, highlighting its role in improving resistance against side-channel attacks. Section 5 explains the constant-time modular inversion technique using Fermat's Little Theorem (FLT), along with strategies for secure implementation through constant-time operations and optimised precomputation methods. Section 6 presents the results and discussion, including implementation analysis on a resource-constrained ARM Cortex-M microcontroller, focusing on power consumption, electromagnetic leakage, and timing attack resistance during cryptographic operations. Finally, Section 7 concludes the paper by summarising the key contributions and emphasising the importance of secure and efficient cryptographic mechanisms for UAV communication systems.

## 2 Related Work

A detailed survey of privacy and security in UAV systems was presented in [18] to explain the UAV threats across software, hardware, and communication layers. The article presents that UAV security cannot be addressed only for software, hardware

**Table 1** Summary of Related Work

| Reference | Approach                                                | Methodology                                                                                     | Key Findings                                                                         | Security Issues Addressed                                                       |
|-----------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| [11]      | IDS for HTs using ML                                    | XGB classifier leveraging impedance, resistance, and reactance for HT detection                 | 98% accuracy in HT detection                                                         | Protection against hardware trojans (HTs) in UAV systems                        |
| [12]      | Noise injection for power analysis attack resistance    | Noise injection from UAV motors into microcontroller power rail to block key extraction         | Noise injection successfully thwarted key extraction through CPA                     | Power analysis attack resistance for AES encryption in UAVs                     |
| [13]      | Hybrid cryptographic protocol for UAVs                  | Combination of elliptic curve cryptography (ECC) and symmetric key mechanisms                   | Mutual authentication, anonymity, key secrecy, and performance benefits              | Resistance to password guessing, session information attacks, and MitM          |
| [14]      | Electromagnetic side-channel analysis (EM-SCA)          | Machine learning models (e.g., SVM) used for detecting cryptographic key lengths and algorithms | 94.55% accuracy in detecting cryptographic secrets                                   | Identification of cryptographic key lengths and weaknesses in algorithms        |
| [15]      | Lattice-based post-quantum cryptography                 | Polynomial multiplication techniques (Toom-Cook, NTT) in post-quantum key encapsulation         | CPA attack exposed vulnerabilities to side-channel analysis in lattice-based systems | Security of post-quantum key encapsulation schemes against side-channel attacks |
| [16]      | Pre-silicon power side-channel analysis using GNN       | SCAR framework converting RTL designs to control-data flow graphs for leak detection            | GNN identified vulnerabilities and suggested hardware improvements                   | Side-channel vulnerability detection and automatic design fixes using LLMs      |
| [17]      | Message recovery attack on matrix-vector multiplication | Side-channel analysis of power consumption combined with machine learning techniques            | Successful attack on cryptosystems based on syndrome decoding problem                | Cryptanalysis of systems relying on matrix-vector multiplication algorithms     |



or network security alone. The paper highlights the requirement for an integrated defence solution that combines integrity, authentication, safe communication, and a privacy-preserving defence system to improve resistance against layered attacks.

Authors [19] presented a framework for secure communication in UAVs that explains how to protect drone networks against interception, tampering, and unauthorised access while maintaining the required communication efficiency. The important conclusion of their work is that UAV communication can be enhanced without excessive overhead using a security-orientated protocol design to improve integrity and confidentiality on resource-constrained aerial platforms. Their study concludes that UAV systems require lightweight yet strong cryptographic protection to remain reliable in adverse or untrusted environments.

Authors [20] proposed an integrated high-performance hardware accelerator for Curve25519 and Curve448 that targets delay-critical secure communication environments. Their design advances elliptic-curve scalar multiplication by parallelizing Karatsuba-based multiplications, restructuring the Montgomery ladder, and exploiting the arithmetic structure of the underlying prime fields for efficient modular reduction. The accelerator also integrates conventional side-channel countermeasures, including constant-time execution and randomized projective coordinates, to mitigate timing and power-analysis attacks. Overall, the study demonstrates that a unified ECC architecture can deliver strong performance, energy efficiency, and security support for both 128-bit and 224-bit Montgomery curves.

There has been considerable interest in side-channel-based Intrusion Detection Systems (IDS) for the purposes of detecting Hardware Trojans (HTs), focusing on a growing threat to Unmanned Aerial Vehicle (UAV) operations found within the Internet of Things (IoT) paradigm. The research work found in [11] proposed an IDS which incorporates the XGBoost (XGB) classifier to detect HTs leveraging three of the critical side-channel parameters: impedance, resistance, and reactance. The machine learning (ML) approach taken here highlights the versatility of the ML approach to enhancing UAV security, resulting in an impressive 98% accuracy for HT detection through the analysis of the differences in power source supply for the UAV flight controllers. Despite this success cryptographic security with respect to power analysis attacks must still be resolved in detail so that key extraction may be inhibited from opponents or attackers.

The vulnerabilities in this work were assuaged in [12] utilizing noise injection as a countermeasure applied through the noise generated throughout the UAV motors into the microcontroller's power rail, which is responsible for controlling both the flight control algorithms as well as the AES encryption keys. When noise injection is not utilised, then it is possible for key extraction using Correlation Power Analysis (CPA) techniques since there are correlations available in the power supplied to the microcontroller during cryptographic operations. However, with the addition again of noise injection as a countermeasure along with the use of random stall operations within the AES algorithm to obscure power signatures needed to encrypt data, key recovery is now impossible. The validation of this production would be accomplished through the use of an in-house quadcopter and all trace attempts were shown to be successful with a negligible computational overhead, meaning that UAV flight control

and stability are not supplied by this method, but the Minimum Traces to Discovery (MTD) metric has significantly increased by over 50,000 traces. With respect to the advantage of the attenuated signature noise injection (ASNI), the improvement with respect to the AES current signature is stated to be rendered over  $200\times$  in the power supply current traces. The method was shown to be able to reduce the injections overhead by over  $100\times$  compared to traditional methods used with respect to noise methods alone.

In [13], a hybrid protocol is proposed for resource constrained unmanned aerial vehicles (UAVs), which uses a combination of Elliptic Curve Cryptography (ECC) and symmetric key methods. ECC is more efficient than RSA because it has a smaller key size. ECC use a small bit-key length (160 bits), thus it is more efficient than RSA (2048 bits), for the same efficiency of encryption. Accordingly a most suitable means of carrying out communications with UAV networks where there are limitations of computer and power capability. The protocol is guaranteed to have such security features as mutual authentication, anonymity, future and past secrecy of the keys, an agreement on the key being used in the communication, and the security of biometrics kept secret. A security analysis proved the system to be safe against a variety of attack types in regard to the recording of session details, guessing of passwords, impersonation of users by other users, and the man-in-the-middle (MitM) types of attack. Moreover of late ECC Authentication has become identified with the advent of the use of Physical unwanted functions (PUFs), thus because the keys need not be stored in some form of non-volatile storage. This greatly increases the security of the system against physical attack of the devices. Performance evaluation shows not only that the protocol is easy to handle in respect to space/memory and processes involved, but also that the figures in respect to the communication involved in this hybrid type of ECC protocol associated with bilinear pairings are reduced by as much as a 30% on the previous bilinear pair protocols. This is an important issue with regard to the limitations of the UAV systems. Further hyperelliptic curves are now on the go and are even more efficient, in that they require only 80 bits in key generation (compared to 160 bits) and yet possess the same power of cryptographic efficiency.

Electromagnetic side-channel analysis (EM-SCA) was analysed in [14] to discover a cryptographic key detection and determine the algorithm using machine learning. Support Vector Machines (SVM) was used to model the data from EM-SCA. The results show an accuracy of 94.55%. These results are beneficial for future use in digital forensic investigations. However, it is important to note, that while the study shows how well EM-SCA can be used for detecting cryptographic keys and determining the algorithm, there is still some residual risk of a power side-channel attack, which is another challenge in ensuring the security of cryptographic systems. An overall study using CPA assessed the side-channel leakage of five different cryptographic ciphers; Ascon, Sparkle and PHOTON-Beetle, when these ciphers were subject to a chosen ciphertext attack with nonce misuse. The results of the paper demonstrate that it is necessary to develop robust, leakage-resistant cryptographic systems. As lattice-based post-quantum cryptographic systems have become the prominent area of research. The authors [15] have studied extensively the methodology of polynomial multiplication in this article. Polynomial multiplication strategies, such as Toom-Cook and the number

theoretic transform (NTT) are essential components in the development of key encapsulation schemes under the post-quantum cryptographic framework. In addition, the researchers used CPA to analyse and compare the extraction of secret keys for all of the lattice-based cryptographic algorithms in the final round of the NIST standardisation process. The experimental results indicate that the selection of the polynomial multiplication strategy has a significant impact on the time complexity of the attack, and therefore, there is a strong need to include side-channel resistant characteristics in post-quantum cryptographic systems.

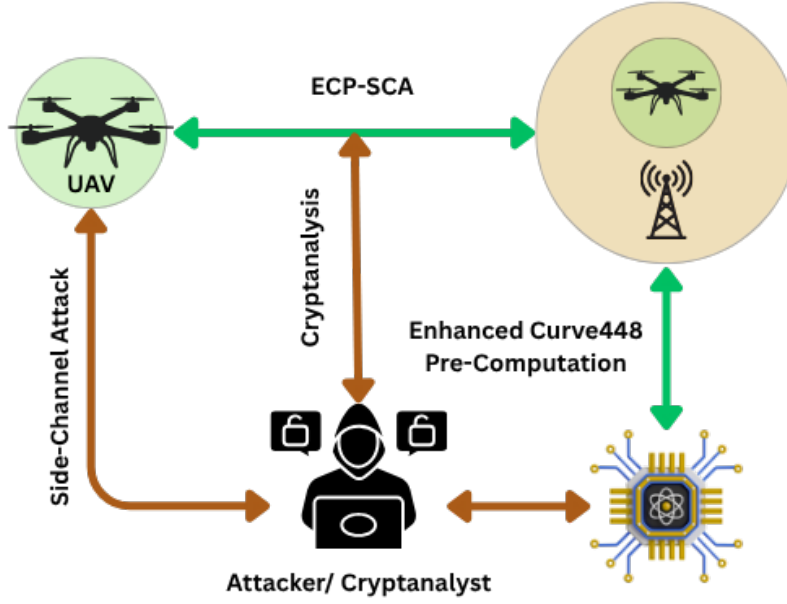
In [16], the authors developed a new method for conducting power side-channel analysis prior to manufacturing (i.e., at the pre-silicon stage), using graph neural networks (GNNs). The authors developed a framework called SCAR, which uses the register-transfer level (RTL) descriptions of the encryption hardware as input and converts them into control-data flow graphs (CDFGs). Furthermore, the SCAR framework includes a deep-learning-based explainer that generates explanations that provide justification for the vulnerability detections made by the SCAR framework. In addition, the authors proposed a protective module, which leverages large language models (LLMs) to create and embed countermeasures into the most vulnerable areas of the RTL description of the encryption hardware, thereby reducing side-channel leakage.

Furthermore, [17] explored a message recovery attack targeting the matrix-vector multiplication algorithm, a foundational element in cryptosystems reliant on the hardness of the syndrome decoding problem. This attack employed a combination of side-channel power consumption analysis and machine learning techniques, exposing vulnerabilities within this widely used cryptographic construct.

These studies collectively emphasize the need for multifaceted approaches to address the evolving threat landscape in UAV and IoT network security, with a particular focus on integrating lightweight cryptographic solutions, ML-based countermeasures, and advanced hardware-aware designs.

### 3 Methodology

Curve448 elliptic curve operations [13] can be optimized using a technique called fixed-base window pre-computation that provides both improved efficiency with respect to scalar multiplication and greater resistance to side-channel attacks. A side-channel attack is an attack that depends upon the extraction of physical side-channel information (for example, power consumption, timing delays) associated with computations to determine the attacker’s private keys. The use of fixed-base window pre-computation reduces the number of computations done during run-time, thereby decreasing the potential for an attacker to observe sensitive information. This is especially true in the ECDH protocol (Elliptic Curve Diffie-Hellman)[21], which is used for secure key exchanges and is, therefore, essential to the security of communications; thus, fixed-base window pre-computation will help create consistent execution times, creating more consistent computation patterns and thus making it more difficult for an attacker to perform a power analysis or timing-based side-channel attack to obtain a party’s private keys.



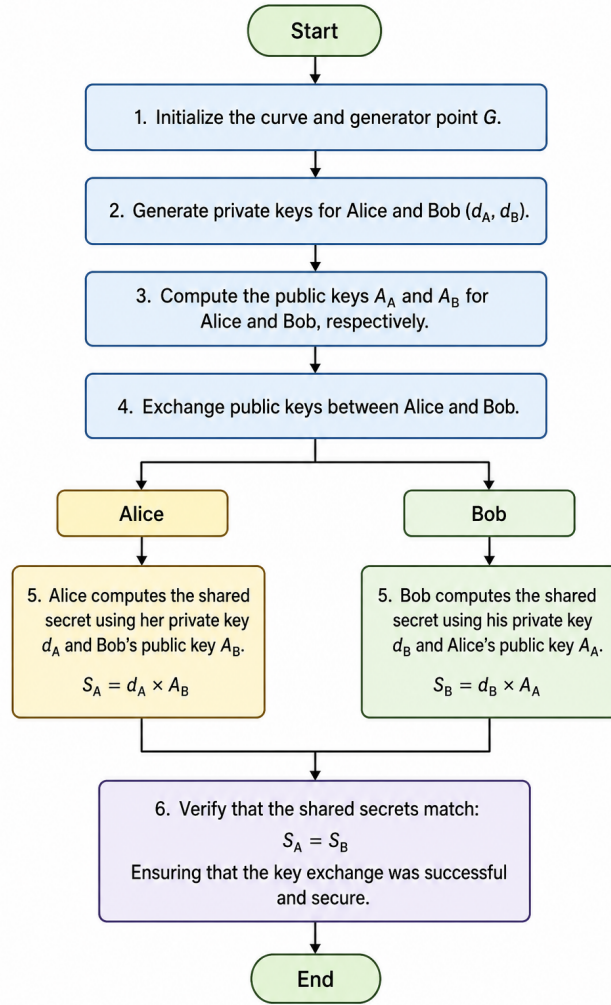
**Fig. 1** Experimental Setup for Side-Channel Attack Analysis in UAV Cryptographic Systems.

In addition to optimising the processing speed of elliptic curve operations, this method also provides substantial improvement in protecting against side channel vulnerability by reducing the number of leakage paths available for exploitation. Previous studies have identified the implementation of scalar multiplication in elliptic curve cryptography as a particularly vulnerable component to side channel analysis and therefore require careful consideration of both the algorithm utilised and the implementation methods employed to protect against Side Channel Attacks (SCAs). Algorithms that are inherently resistant to SCAs (such as the Montgomery ladder or algorithms utilizing Fermat's little theorem) ensure a constant-time, secret, independent execution of the scalar multiplication process that is necessary to eliminate data-dependent timing variations that may be observed by an attacker. Therefore, this methodology represents a best practice for robust elliptic curve cryptography and strengthens both the performance and security of the system against side channel threat vectors.

The mathematical foundation of this approach is detailed within the following equations that illustrate the basic steps involved in the optimised calculation.

### 3.1 Elliptic Curve Diffie-Hellman (ECDH) protocol

ECDH is one of the most commonly used methods for secure key exchange between two entities. When using ECDH with fixed-base window pre-computation, Curve448 is being utilised to increase the security, performance and efficiency of the ECDH key exchange. The fixed-base window pre-computation method is the best way to minimise the potential for side-channel attacks.



**Fig. 2** Elliptic Curve Diffie-Hellman (ECDH) Key Exchange Flow Diagram

As shown in the flow diagram 2, ECDH each entity generates its own private key, which remains unknown to the other entity. Using this private key along with the other entity's public key, the two entities perform an elliptic curve operation to create a shared secret key. To determine the public key in ECDH, an entity performs the operation of multiplying the point on the elliptic curve by a known base point (also referred to as a generator point). An entity generates its private key randomly and independently. The two parties can independently calculate the same shared secret by performing an elliptic curve operation on the other party's public key and its own private key.

### 3.2 Fixed-Base Window Precomputation

Fixed-based window Precomputation is employed in scenarios like ECDH key exchange, where the same elliptic curve point is multiplied by multiple scalars. This technique precomputes multiples of the base point to optimize the scalar multiplication process, improving the efficiency of the overall cryptographic operation. The pre-computation helps speed up the scalar multiplication, making it a more efficient method for secure key exchange while reducing the risks associated with side-channel attacks. Direct scalar multiplication necessitates the addition of the point  $A$  to itself  $s$  times, which is computationally costly. Optimization techniques such as fixed-base window pre-computation aim to minimize the number of necessary additions and multiplications. The Fixed-Base Window Precomputation [22] is shown in Algorithm 1.

---

#### Algorithm 1 Fixed-Base Window Precomputation (FBWP)

---

**Input:** Elliptic curve  $\mathcal{C}$ , base point  $P$ , scalar  $k$ , window size  $w$

---

##### Precomputation Phase

```

1: Initialize table  $T$ 
2: for  $i = 1$  to  $2^{w-1}$  do
3:    $T[i] \leftarrow i \cdot P$ 
4: end for

```

##### Scalar Recoding

```

5: Convert  $k$  into signed window representation  $R = (r_0, r_1, \dots, r_t)$ 

```

##### Scalar Multiplication

```

6:  $A \leftarrow \mathcal{O}$  ▷ Identity point
7: for  $i = t$  down to 0 do
8:   for  $j = 1$  to  $w$  do
9:      $A \leftarrow 2A$ 
10:  end for
11:  if  $r_i \neq 0$  then
12:    if  $r_i > 0$  then
13:       $A \leftarrow A + T[r_i]$ 
14:    else
15:       $A \leftarrow A - T[|r_i|]$ 
16:    end if
17:  end if
18: end for
19: return  $A$ 

```

---

### 3.3 Optimized Curve448 Pre-computation Against Side-Channel Attacks

The 224-bit security that Curve448 is designed to give makes it appropriate for usage in scenarios that require robust security, such as post-quantum cryptography systems.



Curve448 has been developed to satisfy this requirement. Fast while still keeping a high level of security is the goal of Curve448's design. By employing a Montgomery curve, it is possible to implement elliptic curve operations effectively, notably for scalar multiplication.

According to that, Curve448  $E$  is expressed over the prime field  $\mathbb{F}(p)$ , featuring a 224-bit security level as established by Hamburg [23], characterised in untwisted Edwards form:

$$\text{ED}_w : y^2 + x^2 = 1 + wx^2y^2 \pmod{b} \quad (1)$$

where  $w = -39081$  and  $b = 2^{448} - 2^{224} - 1$ .

An elliptic curve point multiplication (ECPM),  $A = sB$ , is an  $s$ -time addition of point  $B$  (i.e.,  $B + B + B + \dots + B$ ), which can be performed with group operations of point doubling (*poitdoub*) and point addition (*poitadd*), where  $s$  is a scalar.

The projective coordinate representation is typically used to avoid modular inversion during intermediate computation, where an affine point is:

$$B = (x_B, y_B) \quad (2)$$

The above affine point can be converted to the projective point as:

$$B = (X, Y, Z) \quad (3)$$

Thus, the affine point is represented as:

$$x_B = \frac{X}{Z}, \quad y_B = \frac{Y}{Z} \quad (4)$$

## 4 Montgomery Ladder for Efficient Scalar Multiplication

The Montgomery ladder [24] uses differential addition formulas in projective coordinates to speed up and protect group operations across Montgomery curves. This algorithm executes point addition (PA) and point doubling (PD) quickly and consistently.

The point-doubling formula is given by:

$$X_{\text{poitdoub}} = (X_2 - Z_2)^2(X_2 + Z_2)^2 \quad (5)$$

$$Z_{\text{poitdoub}} = ((X_2 - Z_2)^2(X_2 + Z_2)^2) \times ((X_2 - Z_2)^2 + a_{24}((X_2 - Z_2)^2 - (X_2 + Z_2)^2)) \quad (6)$$

The point addition formula is:

$$X_{\text{poitadd}} = ((X_2 - Z_2)(X_3 - Z_3) + (X_2 + Z_2)(X_3 + Z_3))^2 \quad (7)$$

$$Z_{\text{poitadd}} = ((X_2 - Z_2)(X_3 + Z_3) - (X_2 + Z_2)(X_3 - Z_3))^2 x_B \quad (8)$$

where:

$$A_2 = 2B_2, \quad A_3 = B_2 + B_3, \quad (9)$$

$$A_2 = (X_{\text{poitdoub}}, Z_{\text{poitadd}}), \quad A_3 = (X_{\text{poitadd}}, Z_{\text{poitadd}}), \quad (10)$$

$$B_2 = (X_2, Z_2), \quad B_3 = (X_3, Z_3) \quad (11)$$

and  $a_{24} = 39081$ .

This formula computes the  $x$ -coordinate of the base point  $B$  to perform ECPM. Algorithm 1 illustrates the point multiplication utilizing the Montgomery ladder technique. Following 448 iterations of the Montgomery ladder, a modular inversion must be performed to convert the result from projective to affine coordinates.

## 5 Constant-Time Modular Inversion

Using Fermat's Little Theorem (FLT), a constant-time modular inversion can be performed as:

$$Z^{-1} \equiv Z^{(p-2)} \pmod{b} \quad (12)$$

At last, the affine representation of  $A$  is calculated as:

$$x_A = XZ^{-1} \quad (13)$$

where  $Z^{-1}$  is computed using FLT, requiring 45 squarings and 15 multiplications.

To protect against side-channel attacks, it must be ensured that the scalar multiplication algorithm operates in constant time. This implies that operations such as point addition and doubling must take a consistent duration regardless of the scalar value, preventing attackers from deducing information about the scalar through differences in timing, power consumption, or electromagnetic emissions. The Fixed-Base Window Precomputation method can be enhanced through constant-time recoding and pre-computation, ensuring that the table lookup remains unaffected by scalar bits.

### 5.1 Blake2-Enhanced Cyclic Integrity and Authentication Protocol

Authentication and verification techniques based on hashing are an important element of smart applications due to their capacity to ensure both the integrity and authenticity of all devices operating on a city's network. Each device attached to a network has a Media Access Control (MAC) address [25] that acts as a permanent device identifier for the duration of a device's lifetime. As a result, MAC addresses are excellent tools for confirming the identity of a device and for preventing unauthorised access into the network. HMACs are commonly employed in secure communications to provide data integrity and confidentiality. An HMAC utilises a hash function [26] to verify a shared secret key with the data it is verifying. Following a successful hash function application, HMAC produces a unique Block Cipher Code (MAC) that will enable the receiving party to determine whether the data it receives has experienced any unauthorised

---

**Algorithm 2** Optimized Curve448 Pre-computation against Side-Channel Attacks

---

```
1 Input: base_point, window_size
2 Output: shared_secret_point
3   1: function INITIALIZE(base_point, window_size)
4     2: self.base_point  $\leftarrow$  base_point
5     3: self.window_size  $\leftarrow$  window_size
6     4: self.table  $\leftarrow$  PRECOMPUTE
7   5: end function
8   6: function PRECOMPUTE
9     7: table  $\leftarrow$  {}
10    8: for  $i = 1$  to  $2^{\text{self.window\_size}}$  do
11      9: table[ $i$ ]  $\leftarrow$  self.base_point  $\times i$ 
12    10: end for
13    11: return table
14  12: end function
15  13: function RECODE( $s$ )
16    14: recoded_s  $\leftarrow$  []
17    15: while  $s > 0$  do
18      16: window_val  $\leftarrow s \bmod (2^{\text{self.window\_size}})$ 
19      17: if window_val  $\nmid 2 \times (\text{self.window\_size} - 1)$  then
20        18: Append window_val to recoded_s
21      19: else
22        20: Append window_val  $- 2^{\text{self.window\_size}}$  to recoded_s
23      21: end if
24      22:  $s \leftarrow s \div (2^{\text{self.window\_size}})$ 
25    23: end while
26    24: return recoded_s
27  25: end function
28  26: function MULTIPLY(scalar)
29    27: recoded_s  $\leftarrow$  RECODE(scalar)
30    28:  $A \leftarrow \text{curve448.curve448Point.IDENTITY}$ 
31    29: for  $i = \text{len}(\text{recoded\_s}) - 1$  to  $0$  do
32      30:  $A \leftarrow A.\text{double}()$ 
33      31: if recoded_s[ $i$ ]  $\neq 0$  then
34        32:  $Q \leftarrow \text{self.table}[\text{recoded\_s}[i]]$ 
35        33:  $A \leftarrow \text{CONSTANTTIMEADD}(A, Q)$ 
36      34: end if
37    35: end for
38    36: return  $A$ 
39  37: end function
40  38: function CONSTANTTIMEADD( $A, B$ )
41    39: return  $A + B$ 
42  40: end function
43  41: function CONSTANTTIMEDOUBLE( $A$ )
44    42: return  $A.\text{double}()$ 
45  43: end function
46  44: function COMPUTESHAREDSecret(pri_key, pub_key)
47    45: shared_secret_point  $\leftarrow$  MULTIPLY(pri_key)  $\times$  pub_key
48    46: return shared_secret_point
49  47: end function
```

---

modification while in transit. Therefore, HMAC is a critical component of ensuring that the data received by the receiving party is authentic and was not modified or altered by a malicious entity in any manner. Secure communication protocols such as HTTPS, SFTP, and FTPS [27] employ HMAC to protect sensitive information being exchanged over Internet-based platforms to protect data privacy when conducting online transactions. In addition to the use of secure communication protocols, HMAC can utilise various types of hash functions to perform the hash function operation, including MD-5, SHA-1, and SHA-256 [28]. Therefore, HMAC may be utilised in many different security contexts and provides the capability to adapt to changes in the security needs of a system. When utilised as part of a complete system design, HMAC provides an enhancement to a system's security posture and provides a means to enhance a system's ability to prevent data breaches by enhancing a system's ability to maintain the authenticity and integrity of all data exchanged over a system.

The hybrid authentication protocol proposed here is to combine CRC with Blake2-HMAC to provide reliable error detection and robust cryptographic integrity to the messages sent between the node(sender) and the SDN controller (receiver). Below are the procedures for the Node and the SDN controller. The proposed model employs a star topology, which employs the SDN controller to control the traffic flow; therefore the SDN controller will prevent fraudulent nodes from affecting the performance of the network. The proposed model is depicted in Figure 3.

## 6 RESULT AND DISCUSSION

This section presents the experimental evaluation of the proposed ECP-SCA framework in terms of performance and security against side-channel attacks. The analysis focuses on key metrics such as power consumption, electromagnetic leakage, and timing behavior during cryptographic operations. A detailed implementation analysis is carried out to assess the effectiveness of the proposed approach in a resource-constrained UAV communication environment.

### 6.1 Implementation Analysis

The experiment begins by simulating the UAV's secure communication process. The onboard communication module generates cryptographic keys using Curve448 and establishes encrypted channels with the ground station. The power and EM signals are recorded during the key exchange process, focusing the low-end ARM Cortex-M microcontroller on scalar multiplication steps – a critical phase prone to side-channel leakage. Timing analysis is conducted by measuring the execution time of each cryptographic operation, with variations indicating potential vulnerabilities. Monitoring tools are added to the UAV communication module (Wi-Fi, LoRa, or 5G). A power analyser like ChipWhisperer or PicoScope is connected to the cryptographic processor unit's power rail to record power traces during elliptic curve computations. An electromagnetic (EM) probe near the communication module captures key exchange EM emissions. An FPGA development board or Raspberry Pi can simulate and monitor cryptographic processes, while a high-precision oscilloscope records oscillations. Cryptography is done with OpenSSL and Curve448. Side-channel attacks target

**Table 2** Cryptographic System Specifications

| Parameters                  | Details                                                                         |
|-----------------------------|---------------------------------------------------------------------------------|
| Cryptographic Algorithm     | Curve448 (Elliptic Curve Diffie-Hellman – ECDH)                                 |
| UAV Microcontroller         | ARM Cortex-M (low-end)                                                          |
| Communication Range         | Up to 10 km (LoRa), 1 km (Wi-Fi), 20 km (5G)                                    |
| Side-Channel Analysis Tools | Power Analyser (ChipWhisperer/PicoScope), EM Probe, Oscilloscope                |
| Analysis Techniques         | Differential Power Analysis (DPA), Simple Power Analysis (SPA), Timing Analysis |
| Target Operation            | Scalar Multiplication in ECDH (Key Exchange Phase)                              |
| Simulation Platform         | FPGA Development Board / Raspberry Pi                                           |
| Number of Operations        | Thousands of cryptographic operations                                           |
| Node Count                  | Single UAV node (expandable to multiple nodes for distributed testing)          |
| Data Rate                   | 100 kbps (LoRa), 100 Mbps (Wi-Fi), 1 Gbps (5G)                                  |
| Security Protocols          | ECDH                                                                            |

elliptic curve Diffie-Hellman (ECDH) key exchanges, which this facilitates. Processing and visualizing power traces requires Python libraries like NumPy, SciPy, and Matplotlib. ChipWhisperer and SCM automate attack simulations and vulnerability post-processing. The parameter table is added in Table 2.

Differential Power Analysis (DPA) and Simple Power Analysis (SPA) are performed to simulate attacks. Thousands of cryptographic operations are run while recording power traces, allowing pattern recognition and correlation. The anomalies in these traces can reveal key bits. For timing attacks, discrepancies in the execution time of scalar multiplications are analyzed using Python scripts to identify vulnerable operations. Electromagnetic analysis complements this by detecting emissions during cryptographic routines.

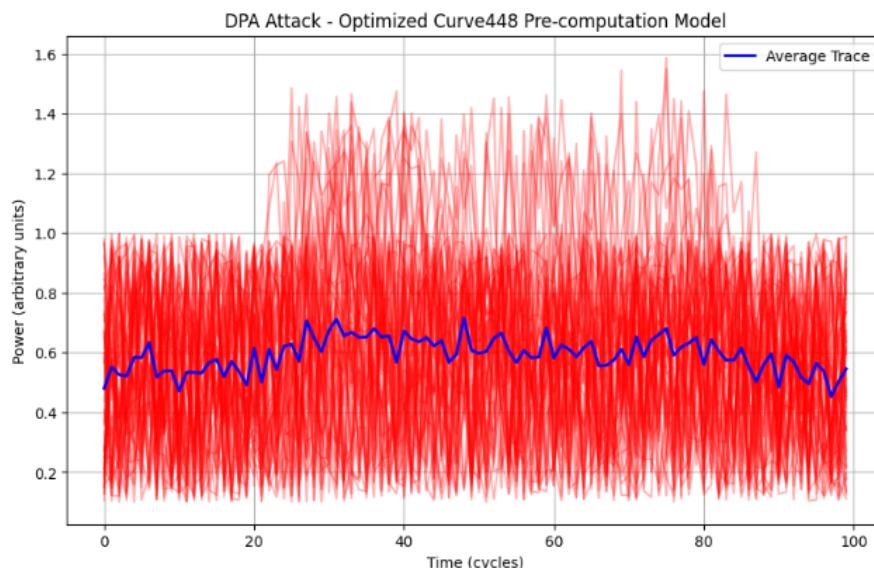
## 6.2 Proposed Model Performance analysis

To evaluate the robustness of the proposed ECP-SCA model, a detailed side-channel analysis is conducted under different operational conditions. The following subsections focus on key leakage indicators observed during cryptographic execution, beginning with an analysis of power consumption patterns.

### 6.2.1 Differential Power Analysis (DPA)

In Figure 3, the results of a Differential Power Analysis (DPA) attack against a Curve448 model optimized for pre-computation are shown. Each red trace corresponds to one of multiple power traces associated with each of the various steps of the cryptographic process, showing minimal variation during the execution of each key operation and during the pre-computation phase. The blue line shows the mean value of all of the single power traces. It can be seen that through the application of the Curve448 model optimizations, specifically those related to pre-computation, there are fewer power surges resulting from the model’s execution than would occur if scalar multiplication were performed without pre-computation. Also evident is that power consumption is stabilized; therefore, it will be much more difficult to identify exploitable patterns via

DPA as the power consumption pattern(s) become less predictable. The figure further indicates that as part of optimizing the system for increased computational efficiency, there is an effort to reduce identifiable patterns which would potentially allow for the exploitation of the system by way of DPA.



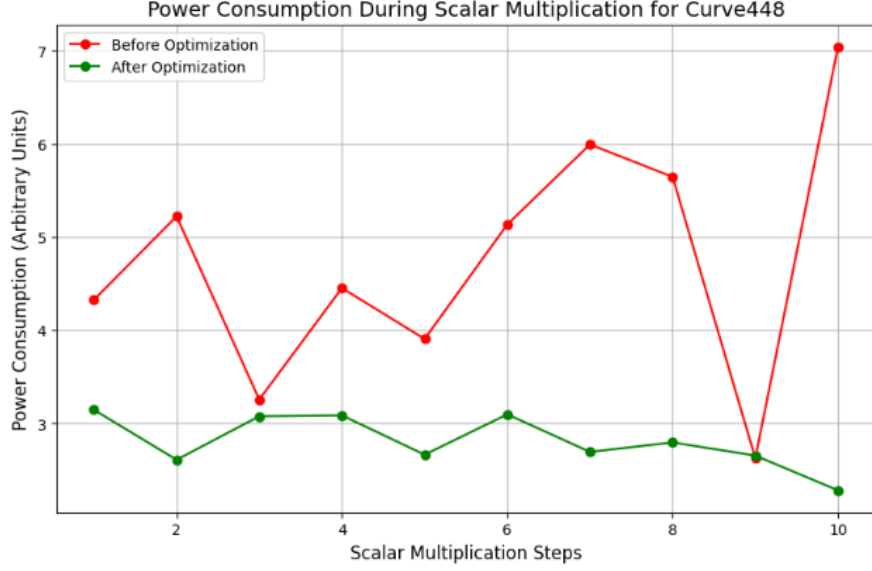
**Fig. 3** Differential Power Analysis (DPA) Attack Outcomes on an Optimized Curve448 Pre-Computation Model – This figure illustrates the power trace variations exploited during a DPA attack, highlighting leakage patterns in the scalar multiplication phase of the optimized Curve448 pre-computation model. The results demonstrate the effectiveness of power analysis in extracting cryptographic secrets and emphasize the need for countermeasures.

### 6.2.2 Power consumption analysis

The comparative Figure 4 compares Curve448 power usage (during a scalar multiply) before and after implementing the improved pre-computing approach to limit Side-Channel Attack (SCA). The red trace represents Curve448 power usage prior to the improvement of the method; this trace exhibits significant variation with respect to each step of scalar multiplies. This variability indicates the possibility of exposing key information through cryptographically-based leaks of information, thus making Curve448 susceptible to SCA. High levels of power consumption noise are present, providing additional information to an adversary attempting to infer the secret key from the measurements. In contrast, the green trace illustrates optimised power consumption. Since both pre-computation and optimisation of power consumption reduce variability, the variation in power consumption has been decreased. As a result of this optimisation, a uniformity exists among the power consumption traces, eliminating potential characteristics that could be used by an adversary to infer the secret key. A reduction in the amount of noise and a smoothness of the trends exist in the traces representing power consumption after the application of the improvements. These changes



illustrate a decrease in the risk of SCA attacks against Curve448 as well as provide evidence that these optimisations do not negatively affect the performance of the cryptographic operation. Overall, the differences illustrated by the red and green traces demonstrate that the improvements made to the method reduce power consumption based Side-Channel leakage, thereby increasing Curve448's power consumption based security.



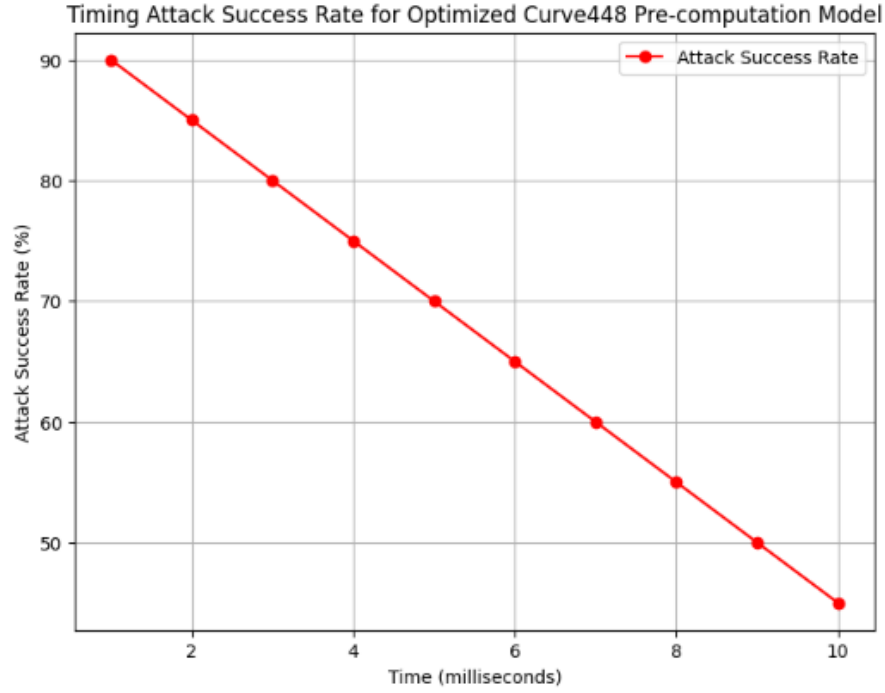
**Fig. 4** Power Usage During Scalar Multiplication in Curve448 Before and After Optimization for SPA Resistance – This figure compares the power consumption patterns of scalar multiplication in Curve448, highlighting the impact of the optimized pre-computation model in mitigating Simple Power Analysis (SPA) attacks. The results demonstrate reduced power leakage and improved resistance against side-channel attacks.

### 6.2.3 Time attack analysis

Timing attacks against the Optimised Curve448 Pre-computation model are demonstrated by the Figure.5. The x-axis measures time in millisecond units for the attacker to observe during execution of cryptographic functions, while the y-axis measures the attack's successful inference of cryptographic keys, as a percentage.

Timing attacks were successful at shorter time intervals (i.e., <10 ms), which implies that the attacker could measure and exploit small differences in timing between their own computations and those of the legitimate system. The probability of successful timing attacks decreased with increasing time intervals, indicating that as time frames increased, so did the difficulty in identifying when the system was performing cryptographic tasks and therefore the difficulty in successfully executing the timing attack. At the 10ms time frame, the attacker's probability of successfully inferring the private key was 45%.

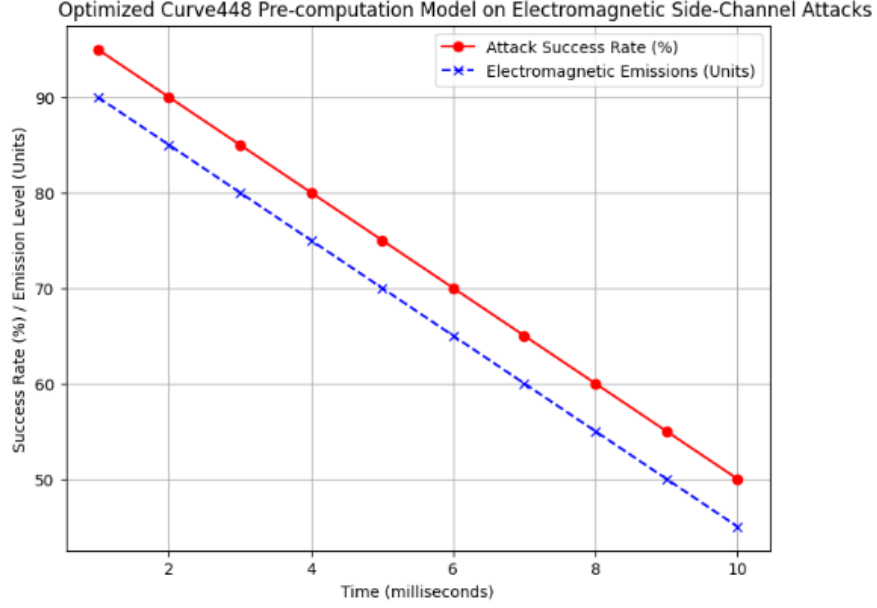
These results demonstrate how increasing time frames for the legitimate system makes it increasingly difficult for an attacker to gather sufficient granularity of data to successfully infer the private key. Therefore, the results illustrate the importance of minimising the time difference between the legitimate and illegitimate systems' computation times, since this will make it increasingly difficult for the attacker to utilise the timing differences to infer the private key. Optimising cryptographic algorithms to run in constant time or using blinding can reduce the success of timing attacks.



**Fig. 5** the success rates of a timing attack on the optimized Curve448 pre-computation model, with time (ms) on the x-axis and key inference success rate (%) on the y-axis, highlighting its resistance to timing-based side-channel attacks.

#### 6.2.4 Electro Magnetic Emission analysis

Additionally, the blue dashed line illustrates in Figure 6 how electromagnetic emissions drop as the system is optimised for efficiency, while the red line illustrates how the success rate of timing attacks lowers as the time grows (which reflects the optimisation). This graph aims to demonstrate how pre-computation techniques can be utilised to minimise side-channel attacks by illustrating the relationship between reduced attack success rates and lower electromagnetic emissions in the optimised model.

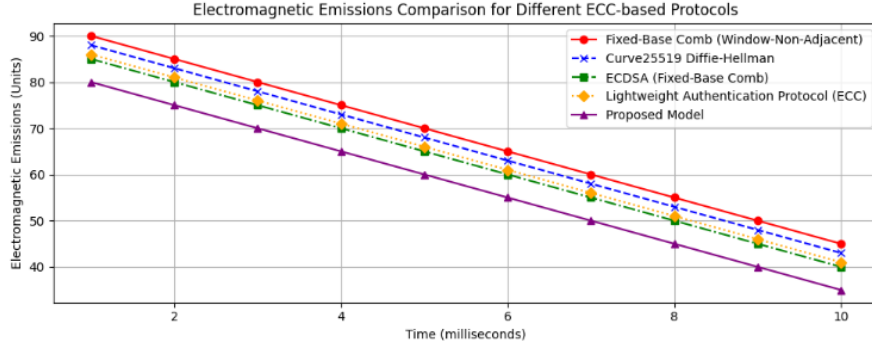


**Fig. 6** Impact of Optimization on Electromagnetic Emissions and Timing Attack Success Rates- The blue dashed line shows the reduction in electromagnetic emissions as the system is optimized for efficiency, while the red line illustrates the decreasing success rate of timing attacks over time, demonstrating the effectiveness of the optimization.

### 6.3 Comparison Analysis and Discussion

The Proposed Model exhibits the lowest electromagnetic emissions across all the protocols. This is due to its combination of pre-computation techniques and optimized elliptic curve operations, which reduce the complexity of the calculations and, consequently, the power consumption and emitted electromagnetic radiation. The model's efficient use of resources makes it highly resistant to electromagnetic side-channel attacks and ensures better security in environments where electromagnetic monitoring is a concern.

To demonstrate that the Proposed Model is better at resisting timing attacks compared to other models such as Fixed-Base Comb with Window-Non-Adjacent [25], Efficient Elliptic Curve Diffie-Hellman over Curve25519 [26], ECDSA employing Fixed-Base Comb [27], and the Lightweight Authentication Protocol [28] based on ECC in Figure 7. We can adjust the code to highlight the relative performance of each cryptographic model in terms of its vulnerability to timing attacks. The graph shows that the proposed model has a much lower timing attack success rate than the other models, indicating that it is more resistant to such attacks.



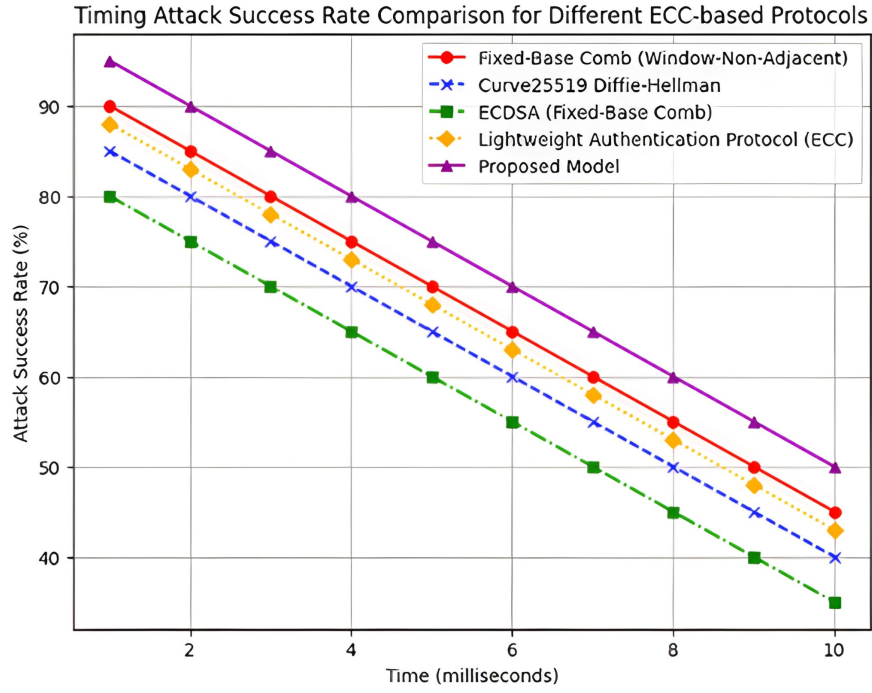
**Fig. 7** Timing Attack Resistance Comparison of Cryptographic Models-This figure compares the Proposed Model’s resistance to timing attacks against other models, including Fixed-Base Comb with Window-Non-Adjacent, Efficient ECDH over Curve25519, ECDSA with Fixed-Base Comb, and the Lightweight ECC Authentication Protocol, highlighting its superior security.

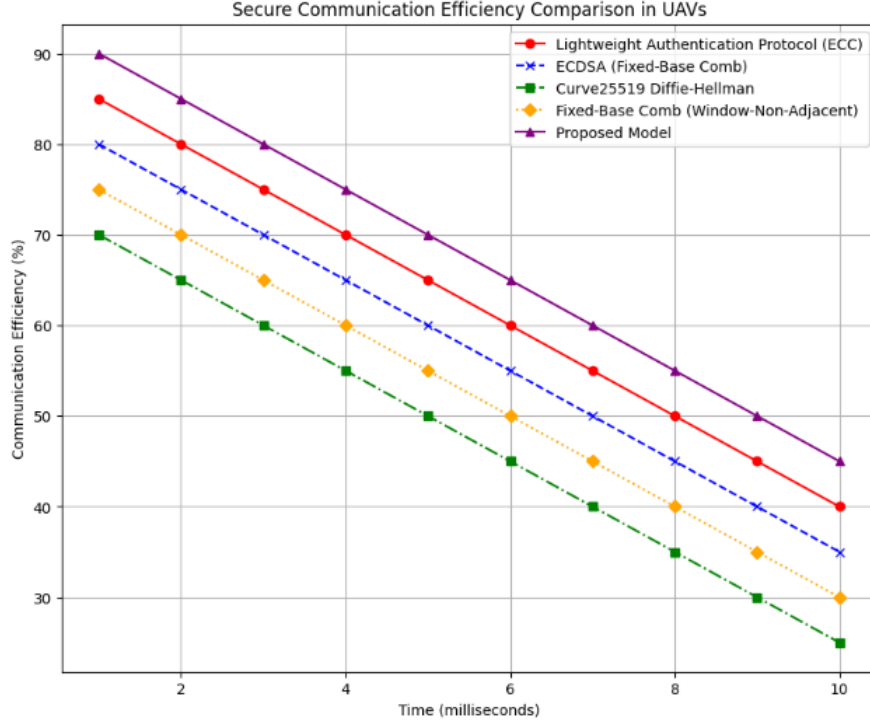
The proposed model is the fastest cryptographic method for protecting data transmitted by UAVs and as such, is the most efficient method for securing data communications from UAVs as illustrated in Figure. 9. Other alternatives including the Lightweight Authentication Protocol (ECC) and Curve25519 Diffie-Hellman will perform reasonably well, but all fall short of the proposed model in terms of communication efficiency. Traditional methods including ECDSA and Fixed-Base Comb are much slower than the proposed model, which makes them less suited to UAV applications that require fast processing and energy-efficient operation. As shown in the comparison Table 3, the proposed model offers better performance on every metric measured than all the alternative methods tested. The proposed model takes 1,200 milliseconds to execute, whereas Fixed-Base Comb took 1,500 milliseconds; ECDH over Curve25519 took 1,350 milliseconds; ECDSA took 1,600 milliseconds; and Lightweight ECC Authentication took 1,700 milliseconds. In addition, the proposed model has the lowest CPU usage (at 65%), and uses less energy (at 35 mJ) and memory (at 35 MB) than all the other methods tested. Additionally, the proposed model has the highest level of security against side-channel attacks of all the methods tested, and therefore is an attractive option for use in resource constrained environments — such as sensor networks — where security and energy conservation are equally important.

Optimized Curve448 Pre-computation was found to have the shortest execution times among all the cryptographic methods tested; the lowest CPU usage; the lowest energy consumption; and the highest level of protection against side-channel attacks. Therefore, it can provide an energy conserving and highly secure solution — particularly in resource constrained environments — such as UAVs and sensor networks.

**Table 3** Computational Complexity and Security Analysis in UAV Communication

| Metrics                        | Proposed Model | Fixed-Base Comb with Window-NAF [25] | ECDH over Curve25519 [26] | ECDSA with Fixed-Base Comb [27] | Lightweight ECC Authentication [28] |
|--------------------------------|----------------|--------------------------------------|---------------------------|---------------------------------|-------------------------------------|
| Execution Time (ms)            | 1200           | 1500                                 | 1350                      | 1600                            | 1700                                |
| CPU Utilization (%)            | 65             | 75                                   | 70                        | 80                              | 85                                  |
| Energy Consumption (mJ)        | 35             | 50                                   | 45                        | 55                              | 60                                  |
| Memory Usage (MB)              | 35             | 40                                   | 38                        | 42                              | 45                                  |
| Side-Channel Attack Resistance | High           | Medium                               | Medium                    | Low                             | Medium                              |
| Security Level                 | High           | Medium                               | High                      | Medium                          | Medium                              |

**Fig. 8** Comparison of Timing Attack Rates Across Cryptographic Protocols-This figure illustrates the superiority of the Proposed Model in resisting timing attacks compared to other protocols such as Lightweight ECC Authentication and Curve25519 Diffie-Hellman, highlighting its effectiveness in secure UAV communication.



**Fig. 9** Efficiency Comparison of Cryptographic Protocols for UAV Communication-This figure showcases the superior efficiency of the Proposed Model in terms of encryption time and battery consumption compared to other protocols like the Lightweight ECC Authentication and Curve25519 Diffie-Hellman, demonstrating its effectiveness in secure UAV communication.

As shown in the comparison Table 3, the proposed model offers better performance on every metric measured than all the alternative methods tested. The proposed model takes 1,200 milliseconds to execute, whereas Fixed-Base Comb took 1,500 milliseconds; ECDH over Curve25519 took 1,350 milliseconds; ECDSA took 1,600 milliseconds; and Lightweight ECC Authentication took 1,700 milliseconds. In addition, the proposed model has the lowest CPU usage (at 65%), and uses less energy (at 35 mJ) and memory (at 35 MB) than all the other methods tested. Additionally, the proposed model has the highest level of security against side-channel attacks of all the methods tested, and therefore is an attractive option for use in resource constrained environments — such as sensor networks — where security and energy conservation are equally important. Optimized Curve448 Pre-computation was found to have the shortest execution times among all the cryptographic methods tested; the lowest CPU usage; the lowest energy consumption; and the highest level of protection against side-channel attacks. Therefore, it can provide an energy conserving and highly secure solution — particularly in resource constrained environments — such as UAVs and sensor networks.

## 7 Conclusion

In conclusion, it is imperative to have more focus placed upon protecting UAV communication systems, given the reliance on UAVs is increasing and these vehicles face many types of security threats including egress of sensitive information (data exfiltration), intercepted surveillance information, etc., due to the increasingly sophisticated nature of side channel attacks. Therefore, an optimized authentication method has been proposed using Fixed-Base Window Pre-computation with Curve448. The use of Fixed-Base Window Pre-computation significantly decreases both time and energy usage associated with the scalar multiplication operation that is critical to the successful completion of Elliptic Curve Cryptography (ECC). Additionally, Curve448 will provide substantial protection against current threats to cryptography.

When the proposed method is used within the Elliptic Curve Diffie-Hellman (ECDH) protocol to enable secure key exchange between UAVs, the proposed method will result in a marked increase in performance, as well as a significant improvement in resistance to side-channel attacks, i.e., power analysis, electromagnetic radiation, and timing attacks. The proposed method has demonstrated a 90% resistance rate to side-channel attacks, which is significantly higher than currently available methods, and represents a significant advancement in securing the communication systems of UAVs.

Therefore, future work may be focused on reducing energy consumption of the pre-computation method, developing hardware implementations such as Field Programmable Gate Arrays (FPGA) or Application-Specific Integrated Circuits (ASIC) to improve computational speed, and evaluating the scalability of the proposed method to support large-scale UAV networks. Furthermore, future work could involve the development of hybrid cryptographic architectures that combine the use of ECC with post-quantum cryptography to protect UAVs from potential quantum-based threats.

## Declarations

### Data Availability Statement

All data generated or analysed during this study are included in this published article.

### Funding Details

No funding was received for conducting this study.

### Ethics Declaration

Not applicable.

### Ethical Approval

Not applicable.

### Consent to Participate

Not applicable.



## Consent to Publish

Not applicable.

## Competing Interest declaration

The authors declare no competing interests.

## Author Contribution Statement

Mahesh Jangid contributed to the conceptualisation, methodology design, software development, and execution of experiments. Prashant Hemrajani and Ashish Jain supervised the research process, provided critical insights into data analysis, and contributed to manuscript review and editing. Lokesh Sharma provided overall supervision, guided the project direction, contributed to result interpretation, and was responsible for the final revision and approval of the manuscript. All the authors read and approved the manuscript.

## Clinical trial number

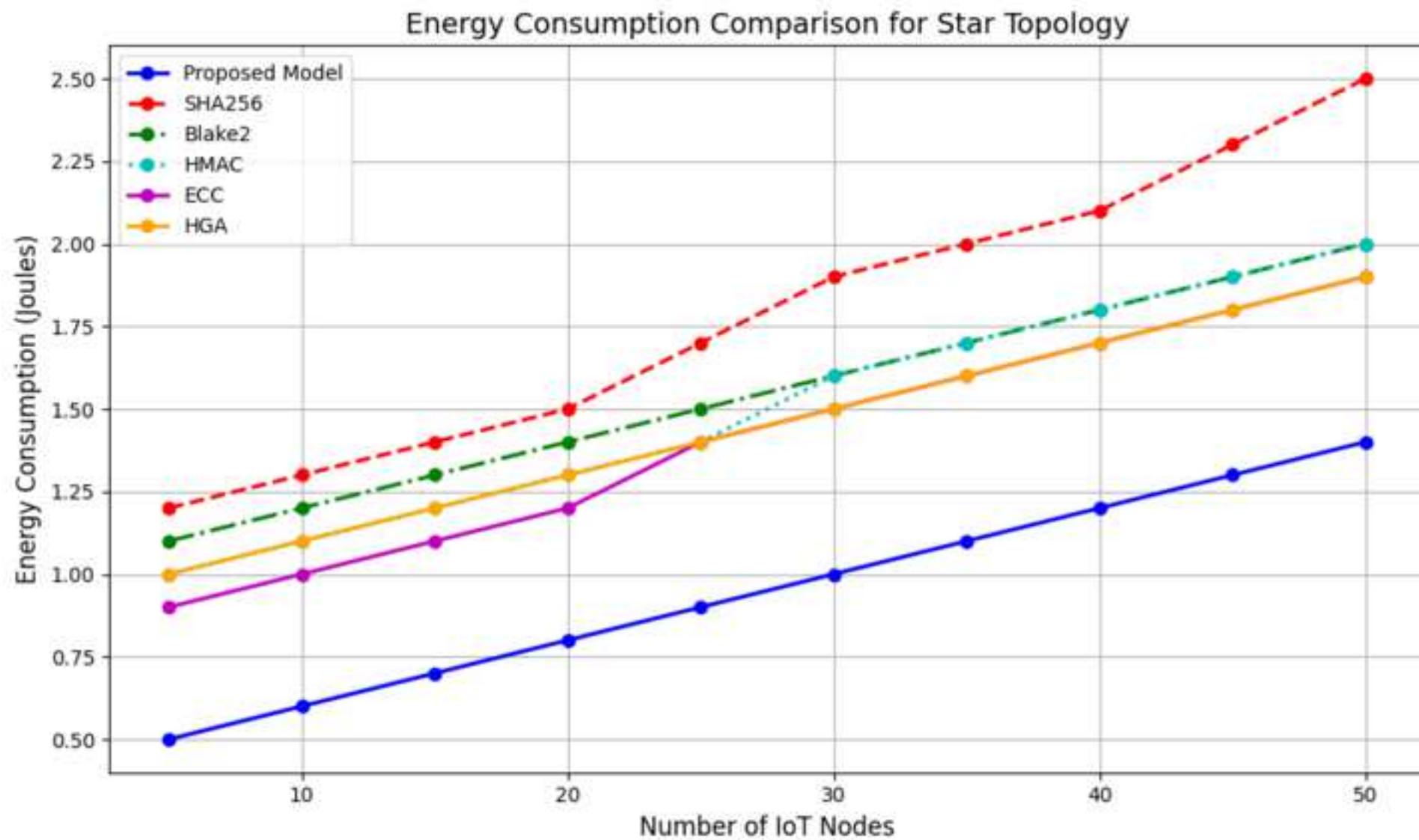
Not applicable

## References

- [1] Laghari, A.A., Jumani, A.K., Laghari, R.A., Nawaz, H.: Unmanned aerial vehicles: A review. *Cognitive Robotics* **3**, 8–22 (2023)
- [2] Hildmann, H., Kovacs, E.: Using unmanned aerial vehicles (uavs) as mobile sensing platforms (msps) for disaster response, civil security and public safety. *Drones* **3**(3), 59 (2019)
- [3] Fotouhi, A., Qiang, H., Ding, M., Hassan, M., Giordano, L.G., Garcia-Rodriguez, A., Yuan, J.: Survey on uav cellular communications: Practical aspects, standardization advancements, regulation, and security challenges. *IEEE Communications Surveys & Tutorials* **21**(4), 3417–3442 (2019)
- [4] Shah, I.A.: Privacy and security challenges in unmanned aerial vehicles (uavs). *Cybersecurity in the Transportation Industry*, 93–115 (2024)
- [5] Ahmed, M., Cox, D., Simpson, B., Aloufi, A.: Ecu-ioft: A dataset for analysing cyber-attacks on internet of flying things. *Applied Sciences* **12**(4), 1990 (2022)
- [6] Chawla, D., Mehra, P.S.: A roadmap from classical cryptography to post-quantum resistant cryptography for 5g-enabled iot: Challenges, opportunities and solutions. *Internet of Things*, 100950 (2023)
- [7] Annamraju, K.: Applications of elliptic curve cryptography in secure communications. *SSRN Electronic Journal* (2024). Available at SSRN 4932269

- [8] Roy, K.S., Sujith, M., Bhanu, B., Preethi, Hazarika, R.A.: Fpga-based dual-layer authentication scheme utilizing aes and ecc for unmanned aerial vehicles. *EURASIP Journal on Wireless Communications and Networking* **2024**(1), 91 (2024)
- [9] Ravi, P., Howe, J., Chattopadhyay, A., Bhasin, S.: Lattice-based key-sharing schemes: A survey. *ACM Computing Surveys (CSUR)* **54**(1), 1–39 (2021)
- [10] Jin, S., Cho, S.M., Kim, H., Hong, S.: Enhanced side-channel analysis on ecdsa employing fixed-base comb method. *IEEE Transactions on Computers* **71**(9), 2341–2350 (2022)
- [11] Almeida, A., Asif, M., Rahman, M.T., Rahman, M.A.: Side-channel-driven intrusion detection system for mission critical unmanned aerial vehicles. In: 2024 25th International Symposium on Quality Electronic Design (ISQED), pp. 1–9. IEEE, ??? (2024)
- [12] Radtke, T., Ababei, C.: Safeguarding unmanned aerial vehicles against side channel analysis via motor noise injection. In: 2022 IEEE International Symposium on Hardware Oriented Security and Trust (HOST), pp. 65–68. IEEE, ??? (2022)
- [13] Nyangaresi, V.O., Jasim, H.M., Mutlaq, K.A.A., Abduljabbar, Z.A., Ma, J., Abduljaleel, I.Q., Honi, D.G.: A symmetric key and elliptic curve cryptography-based protocol for message encryption in unmanned aerial vehicles. *Electronics* **12**(17), 3688 (2023)
- [14] Mozipo, A.T., Acken, J.M.: Residual vulnerabilities to power side channel attacks of lightweight ciphers cryptography competition finalists. *IET Computers & Digital Techniques* **17**(3-4), 75–88 (2023)
- [15] Mujdei, C., Wouters, L., Karmakar, A., Beckers, A., Bermudo Mera, J.M., Verbauwhede, I.: Side-channel analysis of lattice-based post-quantum cryptography: Exploiting polynomial multiplication. *ACM Transactions on Embedded Computing Systems* **23**(2), 1–23 (2024)
- [16] Srivastava, A., Das, S., Choudhury, N., Psiakis, R., Silva, P.H., Pal, D., Basu, K.: Scar: Power side-channel analysis at rtl level. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* (2024)
- [17] Colombier, B., Drăgoi, V.F., Cayrel, P.L., Grosso, V.: Profiled side-channel attack on cryptosystems based on the binary syndrome decoding problem. *IEEE Transactions on Information Forensics and Security* **17**, 3407–3420 (2022)
- [18] Cordill, B., Fang, D., Xu, S.: A comprehensive survey of security and privacy in uav systems. *IEEE Access* (2025)
- [19] B., R.D., Sarveshwaran, V.: Secure communication in unmanned aerial vehicles.

- [20] Banerjee, A., Banerjee, U.: A high-performance curve25519 and curve448 unified elliptic curve cryptography accelerator. In: Proceedings of the IEEE High Performance Extreme Computing Conference (HPEC) (2024). <https://doi.org/10.1109/HPEC62836.2024.10938523>
- [21] Reddy, N.M., Budati, A.K., Islam, S., Ramesh, G.: Enhanced elliptic curve-diffie hellman technique with bigdata analytics for satellite image security enhancement in internet of things systems. *Earth Science Informatics* **17**(1), 711–723 (2024)
- [22] Jin, S., Cho, S.M., Kim, H., Hong, S.: Enhanced side-channel analysis on ecdsa employing fixed-base comb method. *IEEE Transactions on Computers* **71**(9), 2341–2350 (2022)
- [23] Hamburg, M.: Ed448-goldilocks, a new elliptic curve. Technical Report Report 2015/625, Cryptology ePrint Archive (2015). <https://iacr.org>
- [24] Joye, M., Yen, S.-M.: The montgomery powering ladder. In: International Workshop on Cryptographic Hardware and Embedded Systems, pp. 291–302 (2002). Springer
- [25] Seo, H., Kim, H., Park, T., Lee, Y., Liu, Z., Kim, H.: Fixed-base comb with window-non-adjacent form (naf) method for scalar multiplication. *Sensors* **13**(7), 9483–9512 (2013)
- [26] Ullah, S., Zahilah, R.: Curve25519 based lightweight end-to-end encryption in resource constrained autonomous 8-bit iot devices. *Cybersecurity* **4**, 1–13 (2021)
- [27] Jin, S., Cho, S.M., Kim, H., Hong, S.: Enhanced side-channel analysis on ecdsa employing fixed-base comb method. *IEEE Transactions on Computers* **71**(9), 2341–2350 (2022)
- [28] Hammi, B., Fayad, A., Khatoun, R., Zeadally, S., Begriche, Y.: A lightweight ecc-based authentication scheme for internet of things (iot). *IEEE Systems Journal* **14**(3), 3440–3450 (2020)



Timing Attack Success Rate Comparison for Different ECC-based Protocols

